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Dusky Dunes: Blindness Prevention

This quest demonstrated how computer vision is deployed to detect diabetic retinopathy.

The process relied on five core engineering steps:

- **Data Collection:** Gathering thousands of high-resolution retinal fundus images from diverse patient populations to build a robust dataset.
- **Data Labelling:** Enlisting expert ophthalmologists to carefully annotate the images, marking the presence and severity of retinal haemorrhages.
- **Model Training:** Feeding the labelled images into a Deep Convolutional Neural Network (CNN) so it can learn to recognise complex visual patterns of disease.
- **Evaluation & Tuning:** Testing the model against a separate validation dataset to minimise false negatives and optimise clinical accuracy.
- **Deployment:** Integrating the trained AI into accessible screening devices, allowing local healthcare workers to conduct instant, life-saving eye exams.

Market Marshes: Flood Forecasting

This quest illustrated how predictive analytics and physics-informed AI are used to anticipate natural disasters. The workflow followed these five steps:

- **Data Collection:** Aggregating real-time streams of environmental data, including satellite imagery, river gauge levels, and historical rainfall patterns.
- **Terrain Modelling:** Creating highly detailed digital elevation maps to understand exactly how water moves across different landscapes.
- **Model Training:** Training Long Short-Term Memory (LSTM) networks, which excel at processing sequential time-series data to predict future river behavioural patterns.
- **Simulation & Testing:** Running simulated weather anomalies through the model to stress-test its forecasting reliability under extreme conditions.
- **Alert Generation:** Linking the model's real-time outputs to automated public notification systems, sending targeted evacuation alerts directly to at-risk communities.